CONFIDENCE-AWARE PERSONALIZED ADAPTIVE VISUAL ENVIRONMENT CONTROL FOR VR MEDITATION

INDIVIDUAL THESIS

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Individual component: Safety-Constrained Adaptive VR Environment

Group ID: R26-DS-006

Report submitted in partial fulfilment of the requirements for the

B.Sc. (Hons) in Information Technology

Specializing in Data Science

Department of Computer Science

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

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ABSTRACT

Virtual reality meditation can provide a controlled and immersive setting for relaxation, but a fixed visual scene cannot reflect individual preferences or respond safely to changing physiological stress. This research designed and implemented a confidence-aware adaptive visual controller for the Japanese Temple Pond Garden on Meta Quest 2. The component represents the environment through five normalized dimensions: illumination, color warmth, atmospheric softness, color richness, and ambient motion. A participant's long-term and session-specific preferences initialize a safe personalized state. During the adaptive phase, validated stress probabilities, continuous stress, signal quality, preferences, current state, and preference distances form a 24-element context vector. One of three interchangeable study policies then selects an action: Static Personalized, Rule-Based Adaptive, or a disjoint LinUCB contextual bandit. Eleven discrete actions permit only one small directional change at a time, including a first-class NoChange action. A deterministic validator remains authoritative after policy selection and blocks unsafe phase, data, repetition, variation, or transition conditions. Reward is attributed only after a later valid, non-overlapping physiological window and combines normalized stress improvement, RMSSD improvement, heart-rate increase, and visual-change cost. The Unity implementation separates deterministic Core logic from Runtime scene adapters, networking, persistence, and telemetry. Documented EditMode and PlayMode validation covers policy scoring, numerical behavior, reward attribution, session control, safety, and all five Temple Pond mappings. Full Quest 2 endurance, performance, comfort, and human efficacy testing remain future validation steps. The contribution is a transparent, sample-efficient, safety-first learning architecture for personalized visual adaptation in stationary VR meditation.

Keywords: adaptive virtual reality, contextual bandit, LinUCB, physiological stress, personalized meditation, VR safety

ACKNOWLEDGEMENTS

I express my sincere appreciation to my supervisors, for the guidance, critical review, and technical direction provided throughout this research. I also thank the members of research group R26-DS-006 for their work on the mobile, wearable, physiological inference, adaptive audio, and shared integration components that enabled system-level development.

I acknowledge the support of the Sri Lanka Institute of Information Technology and the individuals who assisted with technical discussions and prototype validation. I remain responsible for the design decisions, implementation, interpretation, and conclusions presented in this thesis.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
ADR	Architecture Decision Record
API	Application Programming Interface
HR	Heart Rate
HRV	Heart Rate Variability
JSONL	JSON Lines
LinUCB	Linear Upper Confidence Bound
MVP	Minimum Viable Product
PPG	Photoplethysmography
RMSSD	Root Mean Square of Successive Differences
SDNN	Standard Deviation of Normal-to-Normal Intervals
URP	Universal Render Pipeline
VR	Virtual Reality
XR	Extended Reality

1 INTRODUCTION

This chapter positions the adaptive visual VR component within Laminar VR, defines the individual research problem, states the research gap and questions, and establishes the scope of the contribution. The component changes bounded scene parameters during stationary meditation. It is not a clinical diagnostic system and does not control locomotion, camera motion, or scene switching.

1.1 Overall project context

Laminar VR is a multi-component research platform for personalized immersive meditation. The complete system combines a mobile application, wearable sensing, physiological stress inference, adaptive visual control, adaptive audio control, and Unity-based VR delivery. Shared session and physiology boundaries allow each research component to remain independently testable while contributing to one synchronized meditation session.

Component C owns visual adaptation inside the Japanese Temple Pond Garden. Component B estimates stress and signal confidence from wearable-derived physiology. The mobile application supplies long-term visual preferences and temporary session context. Component C converts those inputs into safe, gradual scene changes and records the evidence needed to evaluate each decision.

1.2 Individual component context

The adaptive visual controller manages five perceptually meaningful variables: illumination, color warmth, atmospheric softness, color richness, and ambient motion. Each variable is normalized to the interval from zero to one before it is mapped to raw Unity properties. This abstraction separates learning logic from the details of one scene and allows future scenes to implement their own calibrated adapters.

Three policies share the same timing, action, safety, transition, reward, and telemetry pipeline. Static Personalized supplies a non-adaptive comparison. Rule-Based Adaptive supplies an explainable population baseline. A disjoint LinUCB contextual bandit learns which safe micro-adjustment is most promising in the present user and environment context.

1.3 Problem statement

A fixed VR meditation scene assumes that the same brightness, warmth, fog, saturation, and motion will remain comfortable and calming for every participant. This assumption ignores

long-term preferences, temporary sensitivity, changing stress, and the possibility that an initially suitable scene becomes less effective during a session.

Directly mapping stress to stronger visual effects is also unsafe. Physiological predictions may be uncertain, delayed, stale, duplicated, or affected by poor signal quality. Visual changes can increase discomfort if they are abrupt, repeated, or applied during invalid lifecycle states. The technical problem is therefore to personalize and adapt the scene while preserving comfort, interpretability, reproducibility, and conservative failure behavior.

1.4 Research gap

Virtual nature and VR meditation research reports promising short-term effects on stress, affect, engagement, and relaxation [1]-[5]. However, many systems present a fixed or researcher-selected environment. Existing adaptive systems often use direct biofeedback mappings or fixed rules, which can make the adaptation visible but do not learn which bounded change is most helpful for a specific participant.

Contextual bandits offer lightweight personalization from sparse interactions [7], [8], while safe learning research emphasizes that exploration must not bypass explicit constraints [9]. Limited work combines long-term visual preferences, probabilistic stress context, signal-quality gating, delayed physiological reward, discrete interpretable scene actions, deterministic VR safety, and deployment on a standalone headset. This integration gap defines Component C.

1.5 Aim and objectives

The aim is to design, implement, and evaluate a personalized adaptive visual controller that uses validated physiological context to make safe and explainable micro-adjustments during stationary VR meditation.

- Represent the meditation environment through five normalized and scene-independent visual dimensions.
- Initialize the scene from participant preferences while retaining a calibrated safe default prior.
- Support Static Personalized, Rule-Based Adaptive, and contextual-bandit study conditions through one control pipeline.
- Use a sample-efficient and interpretable LinUCB learner for sparse session decisions.
- Restrict every proposal through deterministic phase, data, bound, repetition, and variation checks.

- Attribute reward only to valid later physiological evidence rather than current stress alone.
- Integrate mobile preferences, Component B windows, Unity rendering, local telemetry, and session relay behavior.
- Validate deterministic logic and runtime integration before participant-facing Quest deployment.

1.6 Research questions

1. How can visual preferences and scene state be represented independently of raw Unity properties?

2. Why is a contextual bandit suitable for safe personalization under sparse meditation-session feedback?

3. How can learned proposals be prevented from introducing abrupt or uncomfortable visual changes?

4. How can a later physiological response be attributed to one visual action without reusing or overlapping evidence?

5. What implementation and validation evidence is required before the controller can be evaluated on Meta Quest 2 with participants?

1.7 Scope limitations and assumptions

The MVP covers one stationary scene, Japanese Temple Pond Garden, and five bounded visual dimensions. It excludes scene switching, artificial locomotion, camera movement, dense particle effects, near-face motion, flashing, and rapid full-screen changes. The visual controller is independent of the adaptive audio agent, although both may consume the same approved physiology source.

The runtime assumes that Component B supplies timestamped stress probabilities, a continuous stress score, confidence or signal-quality information, HR, and optional HRV measures. The current software evidence does not establish medical benefit. Human comfort, relaxation efficacy, and long-term personalization require ethics-approved participant studies and physical Quest validation.

1.8 Individual contribution

The individual contribution includes the normalized visual domain model, preference initialization, discrete action space, study-policy abstraction, rule baseline, disjoint LinUCB model, feature schema, deterministic safety validator, delayed reward pipeline, stabilization

selector, Unity Temple Pond adapter, configuration profiles, telemetry, persistence support, network-facing boundaries, and EditMode and PlayMode validation.

The Component B prediction model and the Component E audio controller are separate research contributions. Component C consumes approved inputs and coordinates with the shared session path, but it does not alter their internal algorithms or claim their work as part of the visual learning method.

1.9 Thesis structure

Chapter 2 reviews VR relaxation, psychophysiology, contextual bandits, and VR safety. Chapter 3 describes the research and evaluation methodology. Chapters 4 and 5 present the design and implementation. Chapter 6 reports available validation evidence. Chapter 7 discusses the contribution and limitations. Chapter 8 addresses commercialization and sustainability. Chapter 9 concludes and identifies the remaining path to production.

2 LITERATURE REVIEW AND RELATED WORK

This chapter connects evidence from restorative virtual nature, mindfulness, physiological stress measurement, adaptive biofeedback, contextual bandits, safe learning, and cybersickness. The synthesis justifies a conservative, interpretable, and preference-aware visual controller rather than unrestricted online deep reinforcement learning.

2.1 Review method and scope

A structured narrative review was conducted across PubMed, ACM, JMLR, IEEE-related sources, publisher databases, and official Unity, OpenXR, and Meta documentation. Search concepts combined virtual nature, VR meditation, stress recovery, HRV biofeedback, adaptive VR, contextual bandits, LinUCB, safe reinforcement learning, cybersickness, and standalone-headset performance.

Priority was given to peer-reviewed primary studies, systematic reviews, original algorithm papers, recognized physiological guidance, and official platform documentation. Sources were selected when they informed the problem, visual variables, learning method, reward design, safety constraints, human evaluation, or deployment requirements.

2.2 Restorative virtual environments

Valtchanov, Barton, and Ellard reported restorative effects from virtual nature exposure [1]. Annerstedt et al. demonstrated that nature sound combined with a virtual forest supported physiological stress recovery in a pilot study [2]. These findings support coherent natural scenes but do not imply that one fixed rendering configuration is optimal for every user.

Systematic reviews of virtual nature identify generally positive effects on mood, stress, and restorativeness, while also reporting heterogeneous measures, environments, exposure durations, and participant samples [13], [14]. These limitations make personalization and transparent evaluation important. The present research therefore controls a small set of visible scene dimensions and records every change.

2.3 VR meditation and physiological feedback

Modrego-Alarcon et al. found that mindfulness programmes reduced stress and that VR support improved adherence in university students [3]. Jo, Kim, and Lee found reductions in HR, galvanic skin response, and negative affect after both VR and imagery-based guided meditation, with higher concentration reported in VR for some participants [4]. These

outcomes support VR meditation but also show that immersion alone does not guarantee superior stress reduction.

Weibel et al. found that VR-supported HRV biofeedback improved cardiac coherence and user experience in stress management [5]. Biofeedback demonstrates the value of physiological coupling, but direct visual feedback can become distracting if the mapping is too obvious or unstable. Component C instead uses physiology as context for occasional bounded scene decisions.

2.4 Stress HRV and confidence

HR and HRV provide complementary information about autonomic response. RMSSD emphasizes short-term beat-to-beat variability, while SDNN describes broader variability within the analysed window [6]. These measures depend on signal quality, window duration, artifacts, posture, breathing, and individual baseline. They are therefore treated as contextual research signals rather than direct clinical labels.

The controller requires explicit temporal and quality validation before physiology can influence a decision or reward. Signal quality below the configured threshold, stale timestamps, overlapping windows, duplicate windows, or missing required HRV values invalidate learning evidence. This prevents confidence from being treated as a cosmetic dashboard field.

2.5 Contextual bandits and LinUCB

A contextual bandit observes a context, selects one action, receives a reward for that action, and updates its policy. Li et al. introduced a practical LinUCB approach for personalized recommendation, balancing estimated reward with uncertainty [7]. Confidence-bound methods are useful when the learner must exploit current knowledge while allocating limited exploration to uncertain actions [8].

This structure matches the visual MVP better than a large sequential neural policy. The adaptive phase contains only a small number of decision opportunities, each action changes one interpretable dimension, and a later physiology window provides the main outcome. LinUCB learns online with low computation, exposes an action score, supports deterministic testing, and can be reset or persisted explicitly.

2.6 Safe learning and VR comfort

Safe reinforcement learning distinguishes reward optimization from constraints that must hold during learning and deployment [9]. In wellness VR, a high expected reward is not sufficient

if exploration can create a flash, abrupt fog change, excessive motion, or repeated directional drift. Component C therefore places a deterministic validator after policy selection and never allows the learner to bypass it.

Cybersickness research links discomfort to sensory conflict, visual motion, exposure duration, and individual differences [11], [12]. The design avoids artificial locomotion and camera motion, limits ambient motion to gentle pond ripples, uses smooth transitions, and includes local pause and emergency behavior. These choices reduce risk but do not replace physical-headset comfort testing.

2.7 Gap synthesis and conceptual framework

The conceptual framework has five stages: personalize the initial state, validate the current physiology, score available micro-actions, enforce deterministic safety, and attribute a later outcome. The system learns only from valid executed actions. Invalid evidence preserves the current scene and produces no policy update.

Table 2.1 Literature gap synthesis

Evidence area	Established capability	Gap addressed by Component C
Virtual nature	Can support restoration and short-term stress reduction	Most experiences use fixed scene settings
VR meditation	Can improve engagement and concentration	Limited personalized visual adaptation during meditation
Physiological biofeedback	Can connect bodily state with intervention behavior	Direct mappings may be distracting or unstable
Contextual bandits	Provide sample-efficient context-sensitive action selection	Need domain-specific safety and delayed physiological attribution
Safe learning	Separates optimization from non-negotiable constraints	Few examples combine this with bounded restorative VR visuals

3 RESEARCH METHODOLOGY

The research follows a design science approach in which the main output is a tested software artefact and an evaluation protocol. Technical validation is reported separately from human-subject efficacy. This distinction prevents passing software tests from being interpreted as proof that the intervention reduces stress.

3.1 Research design

The work used iterative problem definition, artefact design, implementation, verification, and refinement. Requirements were translated into domain types, policy interfaces, deterministic safety rules, configuration profiles, runtime adapters, telemetry, and automated tests. The same control pipeline supports three study conditions so that comparisons are not confounded by different scene transitions or data validation logic.

The current evaluation consists of code-level tests, Unity runtime tests, scene-wiring inspection, and documented development pairing. A future participant study will compare the three policies using subjective and physiological measures. No uncollected human result is presented as a completed finding.

3.2 Study conditions

Table 3.1 Planned comparable study conditions

Condition	Decision behavior	Learning behavior	Purpose
Static Personalized	Keeps the initialized preference-safe blend	None	Controls for personalization without adaptation
Rule-Based Adaptive	Moves one eligible dimension toward preference when stress or trend rules activate	None	Provides an explainable adaptive baseline
Contextual Bandit	Scores available actions from the current context	Updates executed action after valid delayed reward	Tests lightweight user-specific adaptation

All conditions share preference initialization, physiology validation, the same 11-action vocabulary, safety validation, scene mapping, transition duration, session timing, and telemetry. The policy is the controlled factor. This design supports fairer interpretation than comparing entirely separate implementations.

3.3 Participant and sampling plan

The intended pilot population is adults able to provide informed consent and safely use a Meta Quest 2. Exclusion and stopping criteria should cover relevant motion-sickness history, acute visual discomfort, headache, severe fatigue, photosensitive conditions, inability to understand the procedure, or any clinician-specified contraindication. Final criteria require ethics and supervisor approval.

A pilot sample is appropriate for feasibility, comfort, telemetry completeness, and effect-size estimation. It is not sufficient for broad clinical claims. A later powered study should use a counterbalanced within-participant design or a randomized between-condition design, depending on carryover risk and recruitment capacity.

3.4 Session procedure

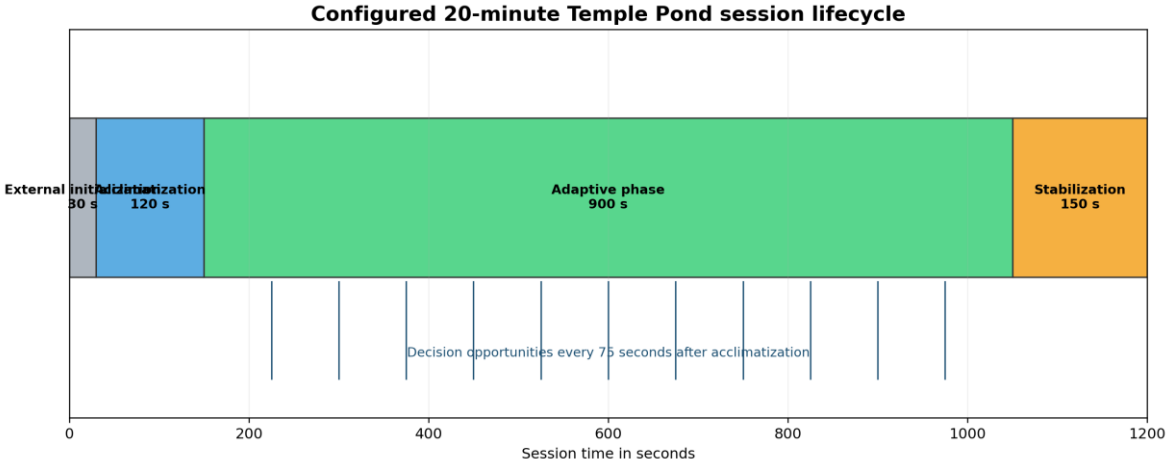


Figure 3.1 Configured Temple Pond session lifecycle

1. Collect informed consent, pseudonymous identity, baseline comfort, long-term preferences, and current session context.

2. Pair the mobile client and Quest runtime, verify the wearable and Component B path, and load the approved configuration.

3. Initialize the scene from the safe-default and effective-preference blend, then complete the 120-second acclimatization period.

4. Run the 900-second adaptive phase with decision opportunities every 75 seconds when valid physiology and safety conditions are present.

5. Stop exploration during the 150-second stabilization phase and select a recent safe state or the blended initial state.

6. Collect post-session stress, visual comfort, simulator-sickness symptoms, personalization ratings, and qualitative feedback.

7. Export local JSONL telemetry and verify session completeness before analysis.

3.5 Measures and instruments

Table 3.2 Proposed evaluation measures

Measure group	Variables	Role
Primary subjective	Pre/post perceived stress and immediate calmness	Estimate participant-perceived change
Physiological	Continuous stress, HR, RMSSD, SDNN, signal quality	Support action-level and session-level analysis
Visual comfort	Eye strain, headache, motion discomfort, distraction	Safety and acceptability
Personalization	Preference fit, naturalness, agency, trust	Assess whether adaptation feels individually appropriate
System behavior	Actions, rejects, transitions, rewards, updates, timeouts	Audit controller decisions
Performance	Frame time, thermal behavior, memory growth, connection events	Quest deployment readiness

The Simulator Sickness Questionnaire can support structured discomfort assessment [15], while a short usability instrument such as the System Usability Scale can support interaction evaluation [16]. Any modified questionnaire must retain clear scoring rules and should not be presented as a validated clinical scale without evidence.

3.6 Data analysis plan

Technical analysis will report test pass status, rejected-input reasons, decision counts, safety modifications, action frequencies, completed reward cycles, missing-data rates, transition durations, and performance traces. Policy comparison will use the same session and safety configuration for all conditions.

A pilot participant analysis should report descriptive statistics, paired pre/post changes, confidence intervals, and effect sizes. With small or non-normal samples, robust or non-parametric tests are preferable. Physiological findings should be interpreted with signal quality, baseline variability, missingness, and multiple-comparison control. Qualitative feedback should be coded for comfort, naturalness, predictability, and perceived personalization.

3.7 Reliability validity and reproducibility

Reliability is supported through versioned configuration IDs, normalized state, deterministic tie handling, finite-value checks, pseudonymous telemetry, and repeatable automated tests. Internal validity is improved by sharing one coordinator and safety pipeline across study policies. External validity remains limited by one scene, one headset class, and the future pilot sample.

Construct validity is protected by separating stress prediction, visual comfort, preference fit, and software success. A lower stress score cannot by itself prove that one visual action caused relaxation. The delayed attribution pipeline therefore requires a valid pre-action window, one executed action, a settling period, and a later non-overlapping window.

3.8 Ethical privacy and safety considerations

Participation must be voluntary, based on informed consent, and interruptible at any time. Emergency stop must remain local and functional without network access. The study must use pseudonymous identifiers, data minimization, retention limits, protected transfer, and controlled access. Physiological data should not be used for diagnosis or shared beyond the approved research purpose.

The system should not encourage participants to remain in VR when uncomfortable. Visual limits, smooth transitions, stationary interaction, explicit exclusion criteria, observation procedures, and post-session checks are required. Adverse events and aborted sessions must be recorded rather than removed from the dataset.

4 SYSTEM ANALYSIS AND DESIGN

This chapter translates the research problem into a safety-first control architecture. It explains the state, action, policy, reward, timing, and failure decisions that determine how the component behaves during a meditation session.

4.1 Functional and non-functional requirements

Table 4.1 Functional and non-functional requirements

ID	Requirement	Type
FR1	Load five normalized long-term and optional session visual preferences	Functional
FR2	Initialize a safe personalized Temple Pond state	Functional
FR3	Receive and validate timestamped Component B physiology windows	Functional
FR4	Select one of eleven interpretable visual actions	Functional
FR5	Apply smooth scene changes through a calibrated adapter	Functional
FR6	Attribute a later physiological outcome to one executed action	Functional
FR7	Record complete decision, safety, reward, lifecycle, and transfer evidence	Functional
NFR1	Fail closed under stale, low-quality, invalid, or missing input	Safety
NFR2	Avoid camera motion, scene switching, flashing, and abrupt visual effects	Safety
NFR3	Keep policy computation small enough for standalone Quest deployment	Performance
NFR4	Separate deterministic domain logic from Unity scene objects	Maintainability
NFR5	Use versioned configuration and pseudonymous local-first telemetry	Reproducibility

4.2 Component architecture

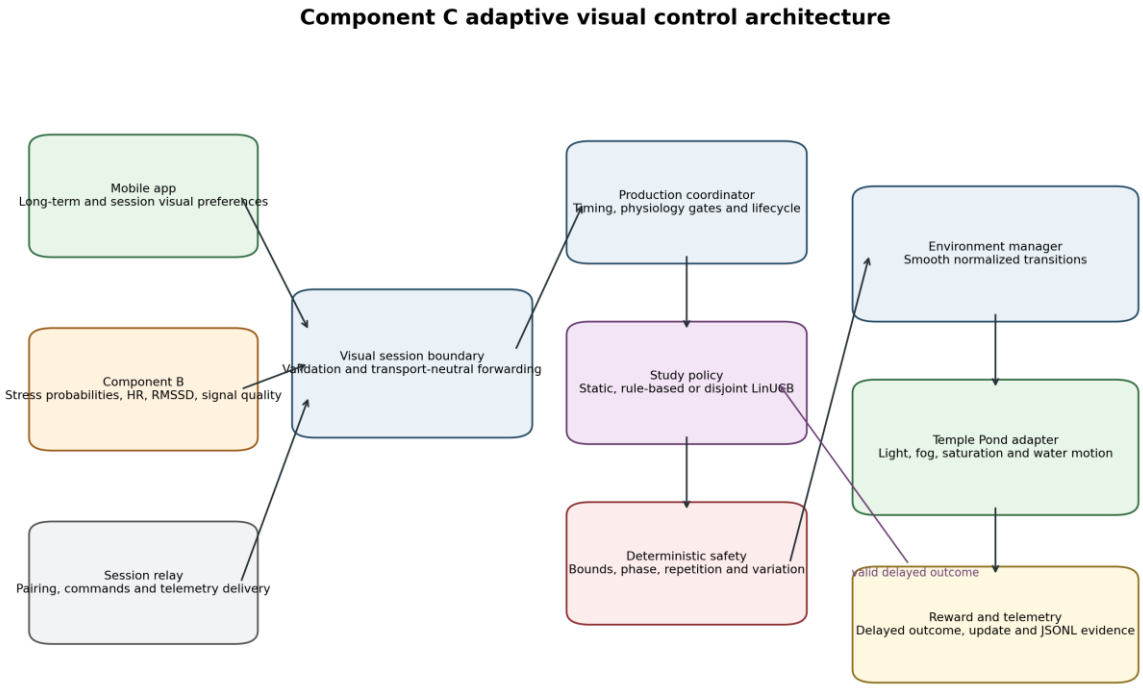


Figure 4.1 Component C architecture within Laminar VR

The VisualSessionBoundary is the transport-neutral ingress. It forwards validated session context, relay connection state, Component B physiology, and commands to the ProductionSessionCoordinator. The coordinator owns phase timing and creates decision opportunities only when the current data and lifecycle state are eligible.

The selected study policy proposes an action from a shared observation. ActionSafetyValidator remains authoritative and can reject or replace the proposal with NoChange. EnvironmentParameterManager performs the normalized transition, and TemplePondEnvironmentAdapter maps the resulting state to explicit scene references. Reward and telemetry are processed without giving the policy direct access to Unity objects.

4.3 Normalized visual state and scene mapping

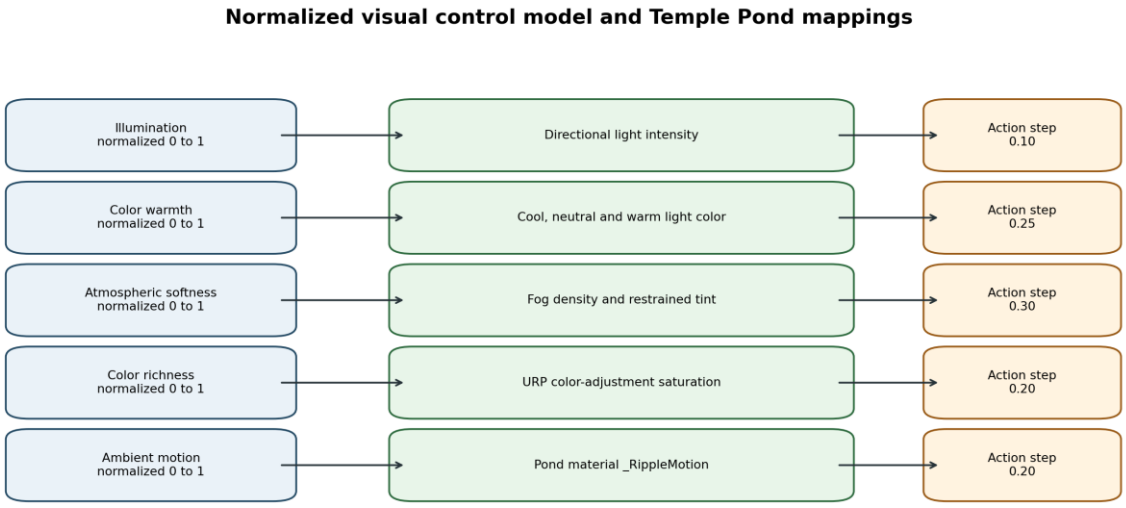


Figure 4.2 Normalized visual controls and Temple Pond mappings

Table 4.2 Visual state and current Temple Pond mapping

Dimension	Normalized meaning	Unity mapping	Safe default
Illumination	Overall light level	Directional light intensity 0.6 to 9.0	0.319
Color warmth	Cool to warm light character	Interpolated cool, neutral, and warm colors	0.500
Atmospheric softness	Clear to softly misted atmosphere	Fog density 0 to 0.015 and restrained color	0.000
Color richness	Muted to richer color	URP saturation -20 to 20	0.500
Ambient motion	Still to gently active pond	Material property _RippleMotion 0.20 to 0.45	0.750

EnvironmentState accepts only finite normalized values and provides L1 and Euclidean distance operations. Normalization makes preference distance, total variation, and action magnitude comparable across dimensions. The adapter uses explicit references and a material property block so the controller does not mutate a shared material asset.

4.4 Preference initialization

The mobile application collects illumination, warmth, atmospheric softness, color richness, and ambient motion as normalized onboarding preferences. Before each session, the participant may keep the usual profile or provide temporary session values. Temporary values affect only the current session and do not silently overwrite the long-term profile.

PreferenceInitializer blends the effective preference with the Temple Pond safe default using a versioned configuration. This compromise avoids beginning at an extreme preference value before the participant has acclimatized. The same initialized state is used by all study policies so that condition comparisons start from an equivalent personalized scene.

4.5 Action space

Table 4.3 Eleven-action discrete visual action space

Action group	Actions	Configured step
No change	NoChange	0
Illumination	IncreaseIllumination, DecreaseIllumination	0.10
Warmth	IncreaseWarmth, DecreaseWarmth	0.25
Atmospheric softness	IncreaseAtmosphericSoftness, DecreaseAtmosphericSoftness	0.30
Color richness	IncreaseColorRichness, DecreaseColorRichness	0.20
Ambient motion	IncreaseAmbientMotion, DecreaseAmbientMotion	0.20

Only one dimension changes in one action. This makes decisions explainable, allows strict consistency checks, and reduces the chance of a compound visual change becoming distracting. NoChange is a first-class action, not an error state, because preserving a good environment can be the correct decision.

4.6 Policy feature design

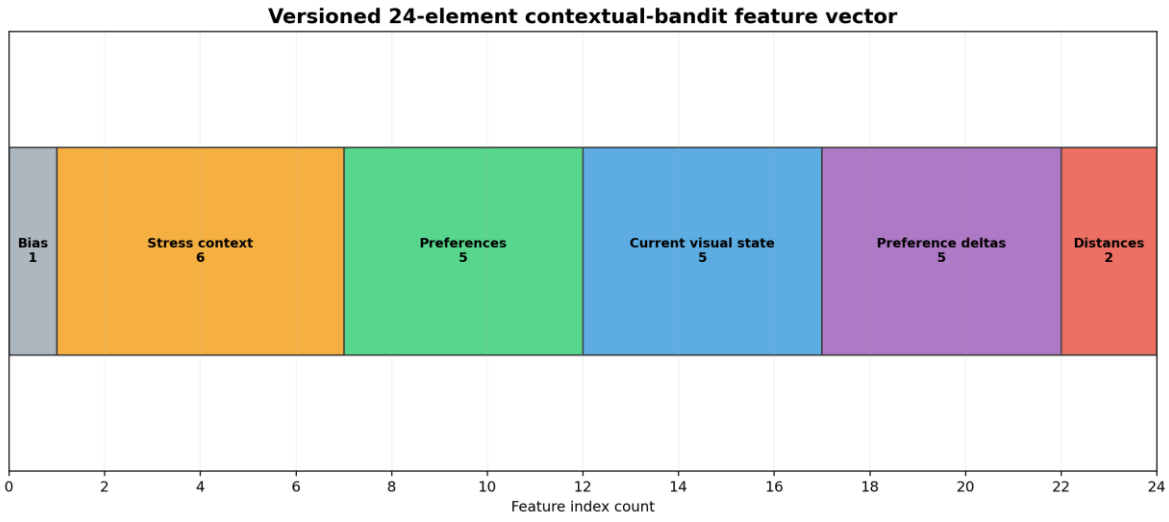


Figure 4.3 Grouping of the 24-element contextual feature vector

The feature vector contains one bias value; normalized continuous stress; four stress-level probabilities; signal quality; five preference values; five current visual values; five current-minus-preference deltas; distance from preference; and distance from the safe default. The exact order is enforced by PolicyFeatureVectorBuilder and tested.

The current code identifies this schema as adaptive-vr-policy-features/0.1-draft. This is an important reproducibility boundary. A participant pilot should not reuse models across schema changes, and the schema must be formally frozen before cross-session persistence is enabled.

4.7 Policy alternatives and selection rationale

Table 4.4 Policy alternatives and design decision

Approach	Strength	Limitation in this component	Decision
Static personalized	Maximum predictability and preference-controlled baseline	Cannot react to changing stress	Retained as control condition
Rule-based	Transparent and easy to validate	Cannot learn participant-specific action effects	Retained as adaptive baseline
Disjoint LinUCB	Fast online learning, uncertainty-aware exploration, interpretable per-action models	Assumes approximately linear immediate reward from context	Selected MVP learner
Deep RL	Can model complex sequential nonlinear behavior	Needs much more data, harder validation, unsafe online exploration risk	Deferred beyond MVP
Direct physiology mapping	Simple visible biofeedback	May oscillate with noise and reveal distracting changes	Rejected as primary controller

LinUCB was selected because one 20-minute session offers only sparse adaptation opportunities. The adaptive phase has at most twelve scheduled opportunities before data gates and reward waiting reduce the number further. A neural policy trained online from this volume would be poorly supported. LinUCB updates one small linear model after each valid outcome and exposes the exploitation and uncertainty terms used in selection.

```
score(a) = x^T theta_a + alpha * sqrt(x^T A_a^-1 x)

theta_a = A_a^-1 b_a
```

The configured model uses ridge regularization 1.0 and exploration coefficient 0.25. Each action has independent A and b state. Only the executed action is updated. Numerical failures,

non-finite values, dimension mismatch, or incompatible snapshots fail safely instead of producing a scene change.

4.8 Deterministic safety design

Candidate filtering removes actions that cannot be executed under current bounds or runtime conditions. Final safety validation then checks the selected proposal again. This second check is required because candidate availability is an optimization, while the validator is the protective authority.

- Adapt only during an eligible adaptive session phase.
- Require network, physiology freshness, signal quality, and non-transition conditions defined by the coordinator.
- Allow only one supported action and one changed dimension.
- Clamp the target to the scene's normalized limits.
- Permit at most two consecutive actions in the same direction.
- Limit total normalized L1 variation across the session to 0.90.
- Use smooth 4.12-second transitions rather than abrupt property assignment.
- Return NoChange under policy, configuration, numerical, data, or safety failure.
- Keep local pause and emergency stop independent of network availability.

4.9 Session timing and decision eligibility

The configured session has 30 seconds reserved for external initialization, 120 seconds of acclimatization, 900 seconds of adaptation, and 150 seconds of stabilization. Decision opportunities occur every 75 seconds during the adaptive phase, but an opportunity does not guarantee an action. The system may remain idle when physiology, network, transition, reward, or safety conditions are not satisfied.

Component B is expected to forward approximately one window per 60 seconds. A window becomes stale after 90 seconds, must cover at least 30 seconds, may be at most 2 seconds in the future, and must meet signal quality 0.50 for decisions and rewards. Sequence and timestamp rules prevent duplicates and reused outcomes.

4.10 Delayed reward and attribution

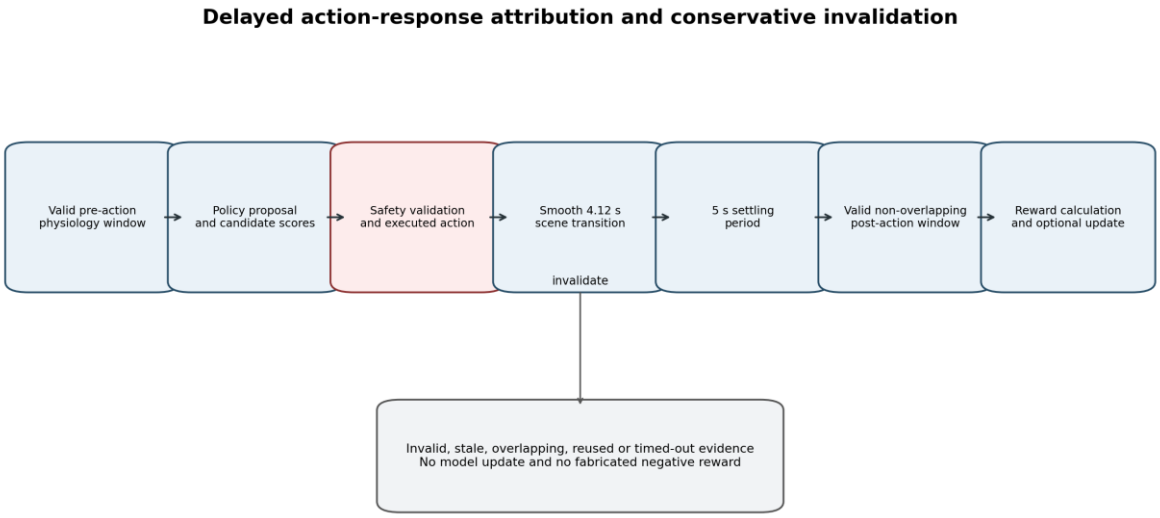


Figure 4.4 Delayed action-response reward attribution

```
reward =
  1.00 zStress + 0.35 zRMSSD - 0.15 zHR
  - 0.10 L1(action) - 2.00 discomfort - 2.00 safety
```

The physiological terms are normalized by participant baseline variability rather than raw units. The pipeline requires at least two baseline samples, a standard-deviation floor of 0.01, a 5-second settling period, and a maximum 120-second wait for an eligible post-action window. Missing RMSSD invalidates a reward when its weight is non-zero.

The current coordinator passes zero for action-aligned discomfort and safety severity because an approved live measurement path is not yet wired. The deterministic safety layer remains active independently. Before a participant trial uses those reward penalties, the system must define when and how discomfort or intervention severity is observed rather than treating missing data as measured zero.

4.11 Stabilization

At the end of adaptation, exploration stops. StabilizationStateSelector considers up to four recent valid outcomes, applies reward recency decay 0.80, and subtracts preference-distance penalty 0.25. Outcomes associated with invalid or unsafe behavior are not eligible. If no suitable candidate exists, the system returns to the blended initial state.

4.12 Standards and professional practice

The implementation applies explicit interfaces, deterministic validation, traceable configuration, fault isolation, and test evidence consistent with software quality principles in

ISO/IEC 25010 [19]. Security planning follows least privilege, pseudonymization, protected transport, minimized identifiers, and auditable evidence aligned with ISO/IEC 27001 [20].

Human-centered design principles from ISO 9241-210 support early preference collection, explicit comfort constraints, interruption control, and iterative evaluation with users [21]. Platform-specific design follows Unity and OpenXR guidance [17], [18]. These standards guide engineering practice but do not certify the prototype.

5 IMPLEMENTATION INTEGRATION AND TESTING

The implementation separates research logic from Unity rendering and networking. This chapter explains the technology stack, assembly boundaries, mobile and Component B integration, Temple Pond mapping, model persistence, telemetry, and test strategy.

5.1 Development environment and technology stack

Table 5.1 Technology stack

Layer	Technology	Purpose
VR runtime	Unity 6000.0.82f1 and C#	Session orchestration, scene control and telemetry
Rendering	URP 17.0.4	Mobile-oriented Temple Pond rendering and saturation control
XR	OpenXR 1.16.1, XR Management 4.5.4, XR Interaction Toolkit 3.0.11	Quest runtime and stationary interaction
Mobile	React Native and TypeScript	Onboarding preferences, session context and pairing
Shared backend	FastAPI, WebSocket and JSONL	Prepared sessions, commands and durable telemetry transfer
Physiology	Component B WebSocket payloads	Stress probabilities, signal quality, HR and HRV windows
Learning	Disjoint LinUCB in deterministic C# Core	Per-action context scoring and online update
Testing	Unity Test Framework 1.6.0	EditMode and PlayMode verification
Target	Android ARM64 IL2CPP on Meta Quest 2	Standalone deployment target

5.2 Core and Runtime separation

The Core assembly contains normalized environment types, policies, LinUCB mathematics, safety, reward attribution, session state, preferences, telemetry contracts, and networking abstractions. It does not depend on UnityEngine scene objects. This makes the research logic deterministic and testable in EditMode.

The Runtime assembly contains MonoBehaviours, ScriptableObject profiles, WebSocket clients, local file stores, the Temple Pond adapter, the production coordinator, pairing interfaces, and main-thread forwarding. Runtime translates between platform events and the Core contracts without changing policy semantics.

5.3 Mobile preference and session context integration

Onboarding stores the five long-term visual preferences and two sensitivity concepts: light sensitivity and motion sensitivity. The session flow records subjective stress, mood valence, fatigue, sleep quality, headache or eye strain, time of day, session sequence, and whether the participant wants usual or temporary visual preferences.

Only the effective five-dimensional preference enters the current visual controller. Additional session context is retained for research and future approved feature design. This prevents unbounded context growth and avoids changing the model schema without a documented normalization and missing-data contract.

5.4 Component B integration

ComponentBWebSocketPredictionSource receives JSON prediction messages and forwards accepted windows through ComponentBPhysiologyBridge. The parser validates required structure, finite numeric values, stress probabilities, timestamps, window order, and signal-quality information before constructing Core physiology objects.

NewestPhysiologyWindowForwardingGate rejects duplicate or out-of-order windows. PhysiologyWindowValidator then enforces duration, future-skew, staleness, probability sum, confidence, and quality constraints. The policy never parses network JSON directly, so a transport change cannot silently change the learning contract.

5.5 Session relay and lifecycle integration

The development session relay prepares one active session, issues a short-lived pairing code, binds the mobile and Quest clients, delivers configuration and commands, and receives visual telemetry. Unity exposes ready, phase, pause, resume, stop, and emergency behavior through a transport-neutral session boundary.

Network reconnection uses bounded exponential backoff. During disconnection, the current safe environment remains rendered, new adaptive work stops, and local telemetry remains available. Production deployment still requires an approved stable schema, WSS or HTTPS, authentication, retention rules, and restart recovery.

5.6 Policy implementation

StudyPolicyFactory creates StaticPersonalizedPolicy, RuleBasedAdaptivePolicy, or ContextualBanditPolicy behind IEnvironmentPolicy. Static policy always returns NoChange.

The rule policy activates from elevated stress or a worsening trend and chooses the largest eligible preference correction. Neither baseline learns from reward.

ContextualBanditPolicy builds the 24-value feature vector, scores eligible actions with DisjointLinUcbModel, and returns the deterministic highest score. The model uses double-precision matrices, ridge initialization, Cholesky-based linear algebra, finite-value checks, deterministic ties, and explicit snapshot metadata. Only an accepted executed action with a valid delayed reward updates its arm.

5.7 Temple Pond scene adapter

TemplePondEnvironmentAdapter applies all five normalized dimensions every transition frame. Illumination sets directional-light intensity. Warmth interpolates calibrated light colors. Atmospheric softness sets fog density and color. Color richness updates the URP Color Adjustments override. Ambient motion writes _RippleMotion through a material property block.

The production build contains JapaneseTemplePondGarden.unity as the only enabled scene. ApplicationBootstrap validates the scene ID and required profile references before creating the environment manager. Invalid or missing research configuration prevents adaptive startup rather than falling back to uncontrolled scene values.

5.8 Configuration and reproducibility

Table 5.2 Current visual research configuration

Configuration	Current identity or value	Research role
Scene	temple-pond	Single MVP environment
Session timing	temple-pond-session-dev-v1	120 s acclimatization, 900 s adaptation, 150 s stabilization
LinUCB	adaptive-vr-linucb-pilot-v1	Ridge 1.0 and alpha 0.25
Physiology	component-b-physiology-pilot-v1	Freshness, quality and window gates
Reward	adaptive-vr-reward-pilot-v1	Baseline, attribution and reward weights
Mapping	temple-pond-mapping-pilot-v3	Raw Unity property ranges
Safety	Coordinator limits	Two same-direction actions and total variation 0.90

Most research-sensitive assets carry explicit IDs, versions, and approval gates. The current TemplePondSceneParameterProfile asset still stores researchConfigurationApproved as false even though ADR-010 and the validation record describe provisional approval. This discrepancy must be resolved and revalidated before participant deployment because fail-closed startup may correctly reject the asset.

5.9 Telemetry and persistence

TelemetryRecorder writes UTF-8 JSON Lines with monotonic sequence numbers and non-decreasing session time. Events cover configuration, physiology acceptance and rejection, policy candidates, proposals, safety validation, transition lifecycle, reward attribution, model update, stabilization, network state, emergency, performance, and completion.

The runtime includes local LinUCB snapshot serialization and compatibility checks. Automatic cross-session learning is not treated as active research behavior until participant identity, feature schema, forgetting strategy, model ownership, consent, and rollback rules are approved. Local telemetry remains the primary reproducibility evidence during network loss.

5.10 Verification strategy

Table 5.3 Verification strategy and evidence boundary

Verification area	Representative checks	Evidence boundary
Environment domain	Normalization, limits, action application and transition state	Deterministic logic only
Policy	Feature order, candidates, scoring, updates, ties, numerical failures and snapshots	Does not prove user benefit
Safety	Phase, network, bounds, repetition, total variation and NoChange fallback	Does not replace Quest comfort testing
Reward	Baseline, overlap, sequence, missing HRV, timeout and update prevention	Does not prove causal physiology
Runtime	Coordinator cycle, pause, emergency, stale input, scene mapping and relay bridge	Editor integration, not full headset endurance
Scene	All five Temple Pond mappings and serialized references	One environment only

5.11 Deployment workflow

1. Start Component B and the development session relay on the approved network host.

2. Open JapaneseTemplePondGarden.unity with the documented Unity 6 editor version.

3. Confirm Android, ARM64, IL2CPP, OpenXR, package identity, and all approved profiles.

4. Prepare a mobile session, obtain the one-time pairing code, and pair the Quest client.

5. Verify valid physiology delivery, scene initialization, readiness, and the delayed start command.

6. Run the session while retaining local telemetry, relay receipts, device logs, and profiler evidence.

7. Abort on discomfort, emergency, invalid configuration, or unsafe runtime behavior.

6 RESULTS AND VALIDATION

This chapter reports the evidence currently available in the repository. It separates verified software behavior from pending device and human evaluation. The current results demonstrate implementation correctness within their tested scope, not clinical effectiveness.

6.1 Static implementation results

- The Core assembly contains the normalized environment, policy, safety, reward, physiology, session, telemetry, and networking abstractions without Unity scene-object dependencies.
- The Runtime assembly contains explicit scene adapters, coordinator behavior, networking, pairing, persistence, and profile conversion.
- The repository contains 92 Core C# files, 49 Runtime C# files, 68 EditMode test files, and 5 PlayMode test files at the analysed commit.
- The current test source contains extensive parameterized EditMode cases and 19 PlayMode UnityTest methods. Annotation counts are not presented as executed test totals.
- Japanese Temple Pond Garden is the only enabled production build scene and contains the visual bootstrap, coordinator, physiology bridge, relay bridge, and scene adapter references.

6.2 Documented automated validation

Table 6.1 Documented software validation results

Validation area	Recorded result	Interpretation
Complete EditMode suite	Reported passing on 29 and 30 August 2026	Supports deterministic domain and integration logic
Adaptive learning cycle	Passed	Bandit updated after a valid delayed reward
Pause and emergency	Passed	Pending work invalidated and transitions frozen
Network and stale physiology	Passed	New decisions froze while current safe state remained
Study policy selection	Passed	All three policies constructed through one factory
Temple Pond mappings	Passed	All five normalized dimensions reached their intended Unity properties
Production coordinator	Passed	Baseline, decision cycle and transport-neutral forwarding executed

Validation area	Recorded result	Interpretation
PlayMode suite	Readiness record reports 17 passing and no failures	Supports Editor runtime integration, not full Quest behavior

The validation record identifies each evidence boundary. EditMode success does not prove scene comfort. PlayMode success does not prove standalone-headset performance. A build would prove compilation but not sustained comfort or thermal behavior. These boundaries are retained in the interpretation.

6.3 Scene and integration results

Unity scene inspection and the validation record show explicit production references for the visual bootstrap, coordinator, boundary, Component B bridge, Temple Pond adapter, and approved profiles. The runtime can initialize scene_id temple-pond and select the configured study policy without giving networking code direct access to the policy internals.

The pilot-readiness record reports development on-device pairing as validated. The relay can receive visual telemetry, publish configured batches, and retain unsent data after a failed send. This supports integration readiness but does not replace a fresh complete mobile-relay-Quest run for the final study build.

6.4 Pending validation

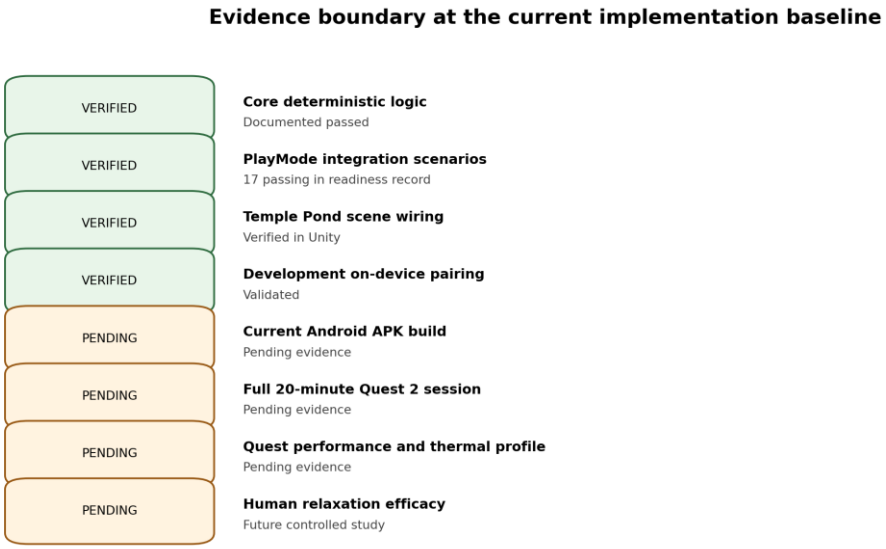


Figure 6.1 Current evidence boundary for Component C

Table 6.2 Remaining evidence required for pilot and production claims

Pending item	Required evidence	Reason
Android development build	New APK path, build log and final package identity	Confirms current player compilation
Full Quest 2 session	Complete 20-minute run with device log	Confirms lifecycle and endurance on target hardware
Performance and thermal	Profiler capture, frame timing, memory and temperature observations	Confirms adaptation does not degrade comfort
Network fault recovery	Wi-Fi loss, reconnect, restart and durable log checks	Confirms fail-closed behavior outside Editor
Visual comfort extrema	Observed transitions across all approved bounds	Confirms calibrated limits on the headset display
Participant efficacy	Ethics-approved comparison with subjective and physiological outcomes	Required for relaxation claims

6.5 Findings by research question

Table 6.3 Findings mapped to the research questions

Research question	Current finding
State representation	Five normalized dimensions successfully separate research logic from raw Unity scene properties.
Learning suitability	Disjoint LinUCB provides uncertainty-aware online personalization with small computation and sparse valid updates.
Safety	A post-policy deterministic validator and NoChange fallback prevent the learner from controlling non-negotiable VR constraints.
Reward attribution	Sequence, overlap, settling, freshness, timeout, and reuse checks prevent invalid windows from updating the model.
Deployment readiness	Editor integration and development pairing are substantially complete, while full Quest and human evidence remain pending.

6.6 Result limitations

The current repository does not contain a completed participant outcome dataset for the visual controller. It also does not retain the machine-readable XML referenced by the readiness summary. The thesis therefore reports the validation record as documented evidence and avoids converting source-code test annotations into a pass count.

The feature schema is still marked draft, the scene parameter approval flag conflicts with the provisional approval narrative, and live discomfort and safety-severity reward inputs are not wired. These are blocking reproducibility tasks before a frozen participant pilot, even though the main control architecture is implemented.

7 DISCUSSION

The results show that safe adaptive VR is a systems problem rather than only an algorithm problem. A useful learner depends on valid physiology, explicit scene semantics, conservative action authority, delayed outcome attribution, calibrated rendering, and evidence that survives network failure.

7.1 Interpretation of the architecture

The five-dimensional normalized state is the main portability decision. It allows the learner to reason about visual intent rather than light-component fields or shader values. A future scene can implement the same dimensions with different raw mappings, provided those mappings are calibrated and validated for that environment.

The discrete one-dimension action space is deliberately less expressive than continuous deep control. That restriction is valuable in a short wellness session because every action can be explained, tested, bounded, and associated with one later outcome. NoChange also allows the policy to preserve a suitable scene without inventing activity.

7.2 Why LinUCB was selected

LinUCB matches the available feedback rate. The learner receives a small number of valid action-response pairs, not millions of simulator steps. It can update immediately from one outcome, quantify uncertainty, and run with negligible computation compared with rendering. Per-action matrices also make model state inspectable and serializable.

PPO, SAC, TD3, and other deep RL methods were not selected for the visual MVP because safe online exploration and participant-specific training data are insufficient. Offline deep pretraining would require a credible visual-response simulator or a large logged dataset that does not yet exist. Using a deep model without that evidence would add complexity rather than justified intelligence.

7.3 Safety as an independent authority

The architecture does not assume that a well-trained policy is automatically safe. Candidate filtering, final validation, smooth transition control, local emergency behavior, and fail-closed configuration remain independent of the reward. This is consistent with safe learning principles and is important because the reward cannot observe every discomfort mechanism.

The strongest limitation is calibration. A normalized bound is only safe when its Unity mapping is comfortable on the target headset. Light intensity up to 9.0, fog up to 0.015, saturation from -20 to 20, and water ripple motion from 0.20 to 0.45 are engineering ranges until physical Quest tests confirm them.

7.4 Comparison with prior work

Prior virtual-nature and VR-meditation studies establish that immersive nature can support restoration and engagement [1]-[4], while VR biofeedback studies show that physiological coupling can influence relaxation behavior [5]. Component C adds a transparent decision layer that combines preference context, probabilistic stress context, uncertainty-aware action selection, and strict visual safety.

Unlike direct biofeedback, the scene does not continuously mirror physiological fluctuation. Unlike a fixed rule system, the contextual bandit can learn different action effects for different contexts. Unlike unrestricted RL, the learner cannot move the camera, switch scenes, combine many dimensions, or override the deterministic validator.

7.5 Individual research contribution

The main research contribution is an end-to-end architecture for bounded adaptive VR visuals under sparse and uncertain physiological feedback. The contribution is not a new contextual-bandit equation. It is the combination of normalized scene semantics, comparable study policies, safety authority, delayed attribution, standalone Unity integration, and reproducible telemetry.

The implementation also contributes practical engineering patterns: a Unity-free Core assembly, explicit ScriptableObject research profiles, safe scene adapters, model snapshot compatibility, local-first JSONL, transport-neutral boundaries, and tests that target failure conditions rather than only successful action execution.

7.6 Limitations

- Only the Japanese Temple Pond Garden is implemented and calibrated for the MVP.
- The feature vector assumes a linear relationship between context and expected action reward.
- The current schema excludes trends, time of day, action history, subjective session context, and additional HRV features.
- Reward attribution remains observational and can be affected by breathing, audio, expectation, posture, and natural stress drift.

- Discomfort and safety-severity values are not yet captured at action level for the reward calculation.
- Cross-session learning infrastructure exists, but an approved forgetting, consent, ownership, and rollback policy is not active.
- The active scene parameter asset has an unresolved approval-flag discrepancy.
- Production authentication, WSS, relay restart recovery, final package identity, and retention procedures remain incomplete.
- No completed human efficacy dataset or full-duration Quest performance evidence is reported.

7.7 Threats to validity

Internal validity may be affected by concurrent audio adaptation, participant expectations, delayed physiology, or imperfect baseline normalization. Construct validity may be affected if continuous stress is treated as a complete representation of relaxation. External validity is limited by one scene, one headset target, and a likely convenience pilot sample.

Implementation validity may be affected by configuration drift between documentation, assets, and deployed builds. The project mitigates this through versioned IDs, approval gates, ADRs, telemetry, and startup validation. The remaining discrepancies identified in Chapter 6 must be closed before the pilot baseline is frozen.

7.8 Future research

Immediate work should freeze the feature schema, resolve asset approvals, wire action-aligned comfort input, build and profile the Quest application, and complete a full fault-injection session. The first human study should emphasize feasibility, naturalness, safety, and telemetry completeness rather than claiming clinical effectiveness.

Later research can evaluate hierarchical context, cross-session forgetting, constrained Bayesian bandits, offline policy evaluation, user-controlled adaptation strength, multiple calibrated scenes, and coordinated audio-visual policies. Any additional feature must have a bounded normalization, missing-data contract, safety interpretation, and version migration plan.

8 COMMERCIALISATION AND SUSTAINABILITY

The component has potential as part of a wellness, education, workplace, or supervised mental-health support platform. Commercial value depends on safety evidence, privacy, reliable deployment, and clear limits on health claims rather than algorithm novelty alone.

8.1 Value proposition and beneficiaries

The value proposition is a VR meditation environment that begins from the participant's visual preferences and changes only through small, explainable, confidence-aware adjustments. Potential beneficiaries include students, employees, wellness users, meditation providers, researchers, and supervised care teams seeking repeatable immersive relaxation sessions.

Organizations may value the audit trail, configurable study policies, local-first operation, and the ability to compare static, rule-based, and learning conditions without rebuilding the scene. These capabilities support research services and controlled pilots before any direct-to-consumer health positioning.

8.2 Commercial model

Table 8.1 Potential commercialization paths

Model	Offering	Requirement
Research platform	Licensed Quest application plus telemetry and policy configuration	Reproducible study package and support
Institutional wellness	Managed deployment for universities or workplaces	Privacy controls, device operations and non-clinical claims
Clinical research support	Configurable visual adaptation for approved studies	Ethics, regulatory review and validated protocols
SDK or middleware	Normalized adaptive-visual controller for third-party scenes	Scene-adapter certification and safety conformance

8.3 Production readiness requirements

- Freeze stable session, physiology, feature, telemetry, and model-snapshot schemas.
- Use WSS or HTTPS, deployment authentication, secret management, access control, and retention procedures.
- Complete repeatable Android build, signing, distribution, rollback, and support workflows.
- Calibrate all visual ranges and transitions on target headsets with diverse users.
- Define incident response, adverse-event handling, emergency behavior, and user support.

- Complete usability, accessibility, comfort, efficacy, and longitudinal personalization studies.
- Avoid diagnostic or treatment claims unless supported by appropriate clinical and regulatory evidence.

8.4 Sustainability

Environmental impact arises from headset manufacture, charging, mobile devices, networking, and data storage. The component reduces unnecessary computation by using a small linear learner, sparse scheduled decisions, local inference, compact JSONL, and one calibrated scene. Local processing also reduces continuous cloud dependency.

Social sustainability requires inclusive preference design, seated and stationary use, interruption control, language accessibility, avoidance of manipulative adaptation, and transparent limits. Economic sustainability depends on maintainable scene adapters, automated regression tests, versioned configuration, and reuse of the same control core across future environments.

8.5 Sustainable Development Goals and responsible innovation

This research supports SDG 3 through accessible stress-management support and SDG 9 through reusable digital infrastructure [22]. These are potential contributions, not proof of population-level impact. Responsible commercialization requires transparent data use, user control, autonomy, and explicit evidence gates.

9 CONCLUSION AND FUTURE WORK

This thesis presented the design and implementation of Component C, a confidence-aware personalized adaptive visual controller for stationary VR meditation. The contribution combines lightweight online learning with explicit VR safety and a reproducible Unity integration path.

9.1 Summary of the work

The environment is represented by illumination, warmth, atmospheric softness, color richness, and ambient motion. Participant preferences initialize a safe blend. During the adaptive phase, validated physiology and scene context form a 24-element feature vector. Static, rule-based, and disjoint LinUCB policies share the same action and lifecycle pipeline.

The learner selects from eleven one-dimension actions. Deterministic safety validates phase, data, limits, repetition, total variation, and transition state. Reward is delayed until a valid later physiology window and is based on stress, RMSSD, HR, and action cost. Invalid evidence cannot update the model.

9.2 Objective achievement

Table 9.1 Objective completion status

Objective	Status	Evidence
Normalized scene-independent visual model	Achieved	EnvironmentState, action steps and Temple Pond adapter
Preference-based safe initialization	Achieved	Mobile preference flow and PreferenceInitializer
Comparable static, rule and learning policies	Achieved	IEnvironmentPolicy and StudyPolicyFactory
Lightweight contextual learner	Achieved	Disjoint LinUCB with tested scoring, update and snapshots
Deterministic safety and smooth transitions	Achieved in software	Validator, limits, transition manager and PlayMode tests
Delayed physiological reward	Achieved in software	Attribution tracker, reward calculator and invalidation tests
Full Quest 2 and participant validation	Pending	Requires APK, profiler, comfort and human-study evidence

9.3 Answer to the research problem

A personalized VR environment can adapt safely by separating learning from authority. The policy estimates which small change may help in the current context, but normalized scene limits, timing, data validation, and deterministic safety decide whether that change is permitted. Delayed reward then updates only the executed action when valid outcome evidence arrives.

9.4 Recommended next steps

1. Resolve the Temple Pond scene-profile approval flag and freeze the 24-feature schema.

2. Wire action-aligned comfort and safety observations, or approve zero reward weights.

3. Rerun all automated suites, retain current reports, and produce a signed development APK.

4. Complete a 20-minute Quest 2 fault-injection and performance session.

5. Run an ethics-approved comparison of all three study policies.

6. Use pilot evidence to calibrate visual ranges, timing, alpha, reward weights, and cross-session learning.

9.5 Final conclusion

Component C establishes a rigorous foundation for conservative adaptive visual meditation. Actions are explainable, unsafe evidence is rejected, and outcomes remain traceable. After Quest and participant validation, it can progress to a production-ready VR subsystem.

REFERENCES

[1] D. Valtchanov, K. R. Barton, and C. Ellard, "Restorative effects of virtual nature settings," Cyberpsychology, Behavior, and Social Networking, vol. 13, no. 5, pp. 503-512, 2010, doi: 10.1089/cyber.2009.0308.

[2] M. Annerstedt et al., "Inducing physiological stress recovery with sounds of nature in a virtual reality forest: Results from a pilot study," Physiology and Behavior, vol. 118, pp. 240-250, 2013, doi: 10.1016/j.physbeh.2013.05.023.

[3] M. Modrego-Alarcon et al., "Efficacy of a mindfulness-based programme with and without virtual reality support to reduce stress in university students: A randomized controlled trial," Behaviour Research and Therapy, vol. 142, 2021, Art. no. 103866, doi: 10.1016/j.brat.2021.103866.

[4] M. Jo, E. Kim, and J. Lee, "Virtual reality vs. imagery: Comparing approaches in guided meditation," Frontiers in Psychology, vol. 15, 2024, Art. no. 1472780, doi: 10.3389/fpsyg.2024.1472780.

[5] R. P. Weibel et al., "Virtual reality-supported biofeedback for stress management: Beneficial effects on heart rate variability and user experience," Computers in Human Behavior, vol. 141, 2023, Art. no. 107607, doi: 10.1016/j.chb.2022.107607.

[6] Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, "Heart rate variability: Standards of measurement, physiological interpretation and clinical use," Circulation, vol. 93, no. 5, pp. 1043-1065, 1996, doi: 10.1161/01.CIR.93.5.1043.

[7] L. Li, W. Chu, J. Langford, and R. E. Schapire, "A contextual-bandit approach to personalized news article recommendation," in Proc. 19th Int. Conf. World Wide Web, 2010, pp. 661-670, doi: 10.1145/1772690.1772758.

[8] P. Auer, "Using confidence bounds for exploitation-exploration trade-offs," Journal of Machine Learning Research, vol. 3, pp. 397-422, 2002.

[9] J. Garcia and F. Fernandez, "A comprehensive survey on safe reinforcement learning," Journal of Machine Learning Research, vol. 16, pp. 1437-1480, 2015.

[10] R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.

[11] L. Rebenitsch and C. Owen, "Review on cybersickness in applications and visual displays," Virtual Reality, vol. 20, pp. 101-125, 2016, doi: 10.1007/s10055-016-0285-9.

[12] S. Weech, S. Kenny, and M. Barnett-Cowan, "Presence and cybersickness in virtual reality are negatively related: A review," Frontiers in Psychology, vol. 10, 2019, Art. no. 158, doi: 10.3389/fpsyg.2019.00158.

[13] S. Riches et al., "Virtual reality relaxation for people with mental health conditions: A systematic review," Social Psychiatry and Psychiatric Epidemiology, vol. 58, pp. 989-1007, 2023, doi: 10.1007/s00127-022-02417-5.

[14] A. Gentile, S. Ficarra, E. Thomas, A. Bianco, and A. Nordstrom, "Nature through virtual reality as a stress-reduction tool: A systematic review," International Journal of Stress Management, vol. 30, no. 4, 2023, doi: 10.1037/str0000300.

[15] R. S. Kennedy, N. E. Lane, K. S. Berbaum, and M. G. Lilienthal, "Simulator Sickness Questionnaire: An enhanced method for quantifying simulator sickness," International Journal of Aviation Psychology, vol. 3, no. 3, pp. 203-220, 1993, doi: 10.1207/s15327108ijap0303_3.

[16] J. Brooke, "SUS: A quick and dirty usability scale," in Usability Evaluation in Industry, P. W. Jordan et al., Eds. London, UK: Taylor and Francis, 1996, pp. 189-194.

[17] Unity Technologies, "Unity Manual, Universal Render Pipeline, XR Interaction Toolkit, and OpenXR documentation," 2026. [Online]. Available: https://docs.unity3d.com/. Accessed: Oct. 11, 2026.

[18] The Khronos Group, "OpenXR Specification," 2026. [Online]. Available: https://registry.khronos.org/OpenXR/. Accessed: Oct. 11, 2026.

[19] ISO/IEC 25010:2023, Systems and software engineering - Systems and software Quality Requirements and Evaluation - Product quality model, International Organization for Standardization, 2023.

[20] ISO/IEC 27001:2022, Information security, cybersecurity and privacy protection - Information security management systems - Requirements, International Organization for Standardization, 2022.

[21] ISO 9241-210:2019, Ergonomics of human-system interaction - Part 210: Human-centred design for interactive systems, International Organization for Standardization, 2019.

[22] United Nations, "Sustainable Development Goal 3: Good Health and Well-Being," 2026. [Online]. Available: https://sdgs.un.org/goals/goal3. Accessed: Oct. 11, 2026.

GLOSSARY

Term	Definition
Action candidate	A currently available visual action presented to the study policy for scoring.
Adaptive phase	Session period during which eligible visual decisions and learning updates may occur.
Contextual bandit	A learner that selects an action from current context and updates from the reward for that action.
Delayed attribution	Association of an executed action with a later valid and non-overlapping outcome window.
Environment state	Five normalized visual values describing the current meditation scene.
LinUCB	A linear contextual-bandit method that combines predicted reward with an uncertainty bonus.
NoChange	An explicit action that preserves the current visual state.
Safe default	Calibrated scene state used as a conservative prior and fallback.
Signal quality	Normalized indication of whether the physiology window is suitable for decisions or rewards.
Stabilization	Final session phase that stops exploration and selects a recent safe state or fallback.

APPENDICES

Appendix A Complete policy feature schema

Table A.1 Current 24-element feature schema

Index	Feature	Range or meaning
0	bias	Constant 1
1	continuous_stress_score_01	Continuous stress divided by 3
2-5	stress_level_0_to_3_probability	Four probabilities from Component B
6	signal_quality	Normalized 0 to 1
7-11	preferred visual dimensions	Five normalized participant preferences
12-16	current visual dimensions	Five normalized scene values
17-21	current_minus_preference deltas	Signed deviation in each dimension
22	distance_from_preference_01	Euclidean distance divided by sqrt(5)
23	distance_from_safe_default_01	Euclidean distance divided by sqrt(5)

Appendix B Core control algorithm

1. Validate session phase, network state, transition state, and latest physiology.

2. Build the current observation from physiology, preference, current state, safe default, and recent controller state.

3. Construct eligible actions, always retaining NoChange where supported.

4. Use the selected study policy to propose one action.

5. Run final deterministic safety validation and derive the executed action.

6. Apply the action to one normalized dimension and begin a smooth transition.

7. Store pre-action state, feature vector, action identity, window identity, and timing.

8. After settling, wait for a fresh valid non-overlapping physiology window.

9. Calculate reward, update only the executed LinUCB arm, and record the full outcome.

10. Invalidate without update on pause, emergency, network loss, timeout, stale evidence, overlap, or duplication.

Appendix C Component B input example


```
{
  "timestamp": 1787282898.4,
  "heartRate": 78.4,
  "rmssd": 34.1,
  "sdnn": 42.0,
  "stress": {
    "mode": "band",
    "level_low": 1,
    "level_high": 2,
    "label": "mild-to-moderate",
    "confidence": 0.54,
    "probabilities": {
      "relaxed": 0.08,
      "mild": 0.36,
      "moderate": 0.54,
      "high": 0.02
    },
    "continuous_score": 1.50
  },
  "signalQuality": 0.92,
  "windowStart": 1787282838.4,
  "windowEnd": 1787282898.4
}
```


Appendix D Quest 2 validation checklist

Table D.1 Physical Quest validation checklist

ID	Scenario	Pass evidence
Q2-01	Full session	No crash, tracking loss, unsafe phase change, or incomplete log
Q2-02	Performance	No repeatable action-correlated frame-time spike and approved numeric budget
Q2-03	Network loss	Current safe scene remains and no new action executes
Q2-04	Emergency stop	Local stop works with transport disconnected
Q2-05	Persistence	Compatible model restores and incompatible identity is rejected
Q2-06	Visual comfort	No flash, abrupt fog, camera motion, near-face motion, or unsafe transition
Q2-07	Pause and resume	Learning freezes and fresh physiology is required after resume
Q2-08	Headset lifecycle	Focus loss or sleep does not fabricate input or resume stale work

Appendix E Research configuration values

Table E.1 Current engineering research values

Parameter	Value	Interpretation
Decision interval	75 s	Scheduled opportunity during the adaptive phase

Parameter	Value	Interpretation
Transition duration	4.12 s	Smooth interpolation of a valid visual action
Settling period	5 s	Delay before post-action reward evidence
Attribution timeout	120 s	Maximum wait for an eligible post-action window
Signal-quality gate	0.50	Minimum for decisions and rewards
Stale threshold	90 s	Maximum age of a physiology window
Repeated-direction limit	2	Maximum consecutive same-direction actions
Total variation limit	0.90 L1	Session-level normalized visual-change budget
LinUCB ridge	1.0	Regularization of each action model
LinUCB alpha	0.25	Uncertainty bonus coefficient

Appendix F Academic integrity and AI use disclosure

Generative AI tools, including OpenAI Codex, were used to assist with document structuring, language refinement, codebase summarization, and formatting. The student is responsible for reviewing all generated text, confirming technical claims against the implementation, verifying references, and making the final submission decision.

The adaptive visual policy is implemented in the research artefact and is not simulated by a language model. The contextual bandit, safety validator, reward pipeline, scene mapping, and telemetry are deterministic C# components in the Unity project.